RPLIDAR C1

From Waveshare Wiki
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Overview

RPLIDAR C1
TTL, UART

Introduction

The RPLIDAR C1 has a ranging radius of up to 12m, with a ranging blind zone as low as 0.05m, making it capable of easily scanning and ranging objects at various distances, addressing challenges posed by various obstacles in the environment. It is suitable for practical applications such as robotic vacuum cleaners, educational ROS (Robot Operating System) cars, commercial robots, low-speed unmanned vehicles in parks, and parking space detection.
The typical scanning frequency of the RPLIDAR C1 is 10Hz (600rpm). At this frequency, a sampling rate of 5,000 can achieve an angular resolution of 0.72°.
The RPLIDAR C1 integrates years of technical expertise in triangulation and Time-of-Flight (TOF) ranging, providing depth fusion. It meets the requirements for robot positioning, mapping, navigation, and offers performance in ranging accuracy, distance measurement, anti-interference, and anti-tailing.
The RPLIDAR C1 consists of a laser-ranging core and the power supply and mechanical components responsible for the high-speed rotation of the laser-ranging core. During normal operation, the ranging core starts rotating clockwise for scanning. Users can obtain the scanning ranging data of the RPLIDAR C1 through its communication interfaces, such as serial port or USB.

Specification

- Next-generation integrated ranging technology
- Supports UART communication
- Measurement radius of 12m
- Measurement frequency of 5000Hz
- 360° scanning range

- Laser safety compliant with Class 1 standards

Parameters

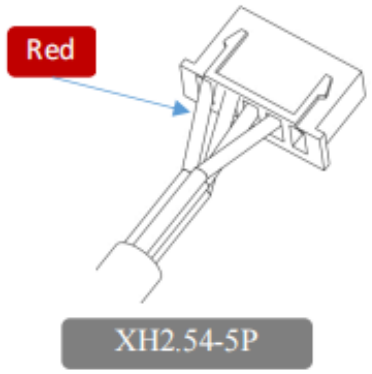
RANGING DISTANCE	0.05~12.00m (white target, 70% reflectivity); 0.05~6.00m (black target, 10% reflectivity)
RANGING FREQUENCY	5000 Hz
SCANNING FREQUENCY	8~12 Hz (Typ. 10 Hz)
ANGULAR RESOLUTION	0.72°
DIMENSIONS (L × W × H)	55.6*55.6*41.3
COMMUNICATION INTERFACE	UART @ 460800bps
POWER SUPPLY	5V
OPERATING CURRENT	230mA
POWER CONSUMPTION	1.15W
OPERATING TEMPERATURE	-10℃~40℃
SCANNING RANGE	360°
RANGING ACCURACY	±30mm

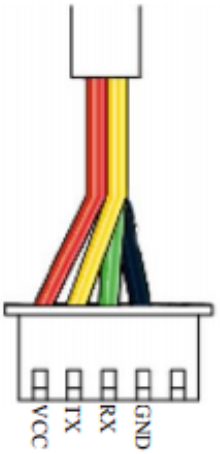
Application Scenarios

- Education and Research
- Robot Obstacle Avoidance
- Measurement and Detection
- Intelligent Gesture Control
- 2D Gesture Recognition
- Autonomous Navigation
- Map Reconstruction
- Navigation and Localization

Working Mode

The standard configuration of RPLIDAR C1 utilizes a 3.3V level serial port (UART) as the communication interface and can, based on customer requirements, adopt other interfaces such as USB. The table below provides specifications based on the serial interface:





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Item	Unit	Min.	Typ.	Max.	Note
Baud Rate	bps	-	460800*	-	-
Working Mode	-	-	8-bit data, 1-bit stop bit, no parity	-	8n1
Output high-level voltage	V	2.9	3.3	3.5	Output signal high level voltage value
Output low-level voltage	V	-	-	0.4	Output signal low level voltage value
Input high-level voltage	V	2.4	3.3	3.5	Input signal high level voltage value
Input low-level voltage	V	0	-	0.4	Input signal low level voltage value

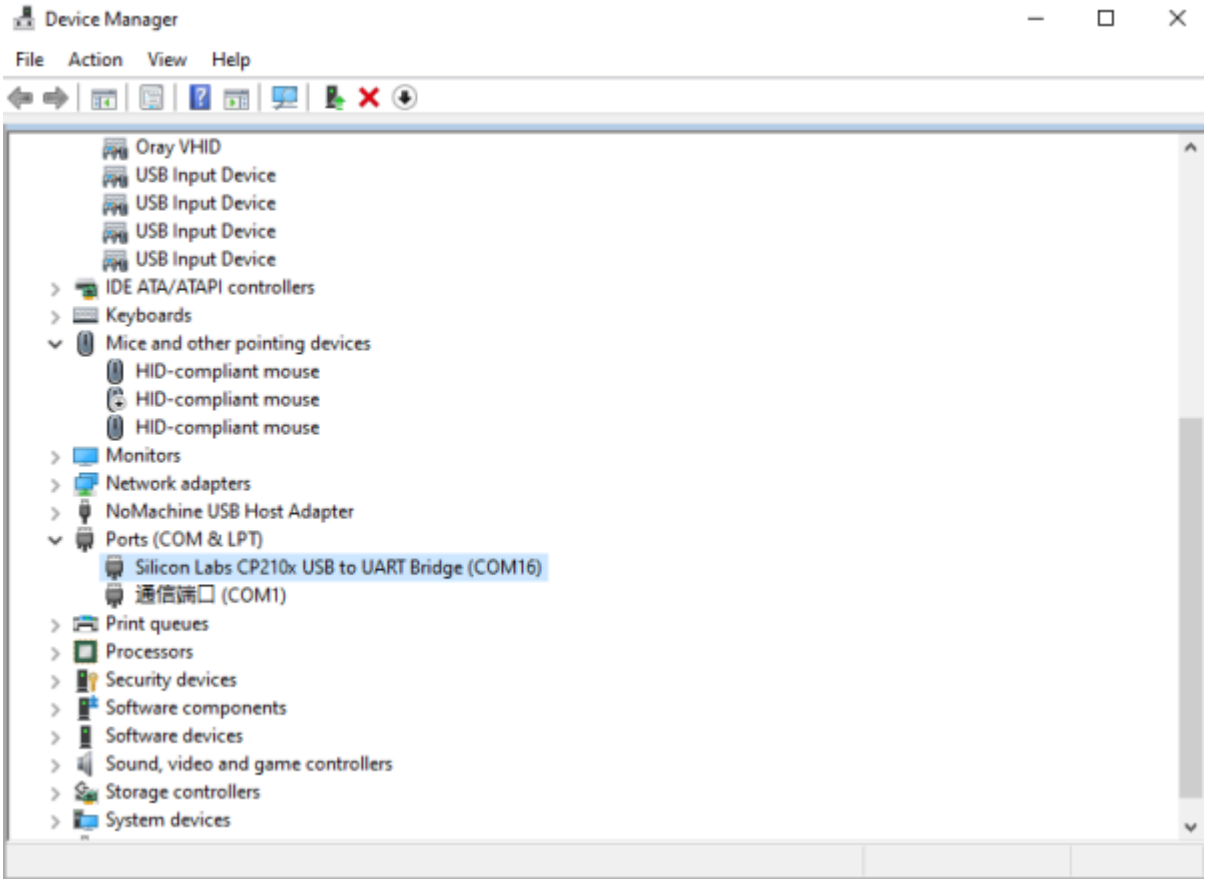
The RPLIDAR C1 uses a separate 5V DC power supply to power both the ranging system and the motor system. The standard RPLIDAR C1 utilizes the XH2.54-5P male socket as defined in the table below:

Signal	Type	Description	Min.	Typ.	Max.
VCC	Power supply	Total power	4.8V	5V	5.2V
TX	Output	RPLIDAR Distance Measuring Core Serial Outputs	0V	-	3.5V
RX	Input	RPLIDAR Distance Measuring Core Serial Inputs	0V	-	3.5V
GND	Power supply	ground	0V	0V	0V

Host Software

- robostudio.zip (https://files.waveshare.com/wiki/RPLIDAR-C1/Robostudio.zip)
- CP210x_Driver.zip (https://files.waveshare.com/upload/6/63/CP210x_Universal_Windows_Driver.zip)
- Connect the LiDAR to the adapter board, and connect the adapter board to the computer through USB.

- Open the device manager, and check the COM port of the CP210x.



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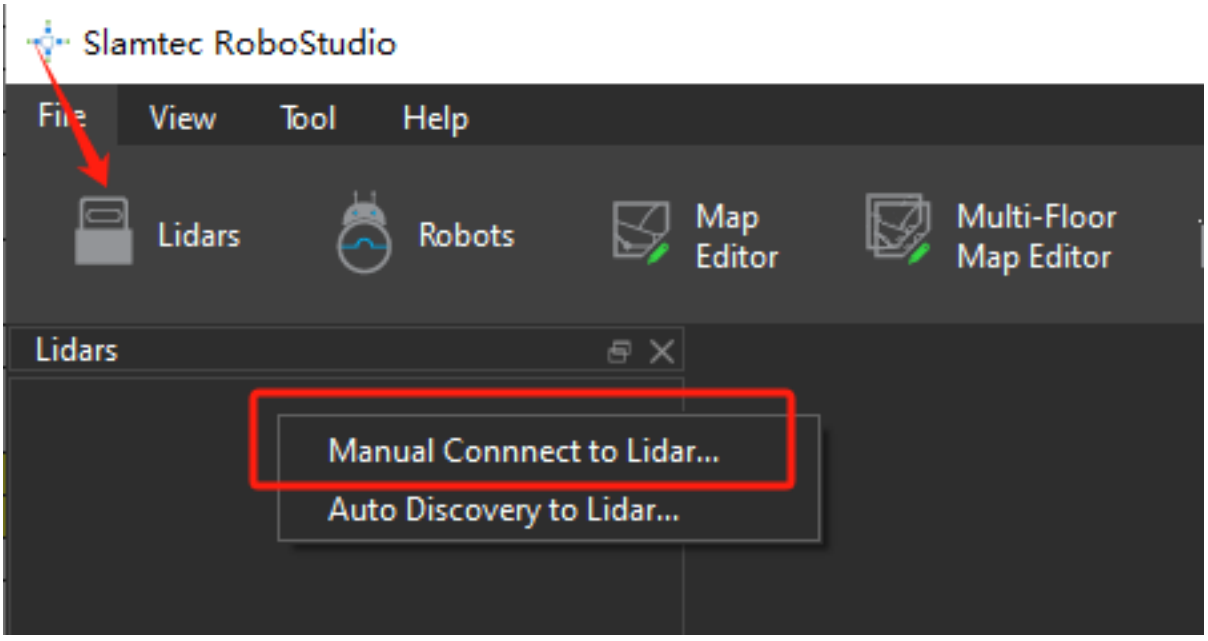
- If the device is not detected then you need to download CP210x_Driver.zip and install the driver.
- Download robostudio.zip and install robostudio, run RoboStudio.exe after installation, register the account and log in.
- Open the RoboStudio interface, click on "File" -> "Lidars", and pop up the radar floating window on the left side. Right-clicking within this window presents two options. Choose "Manual Connect to Lidar" from the menu.



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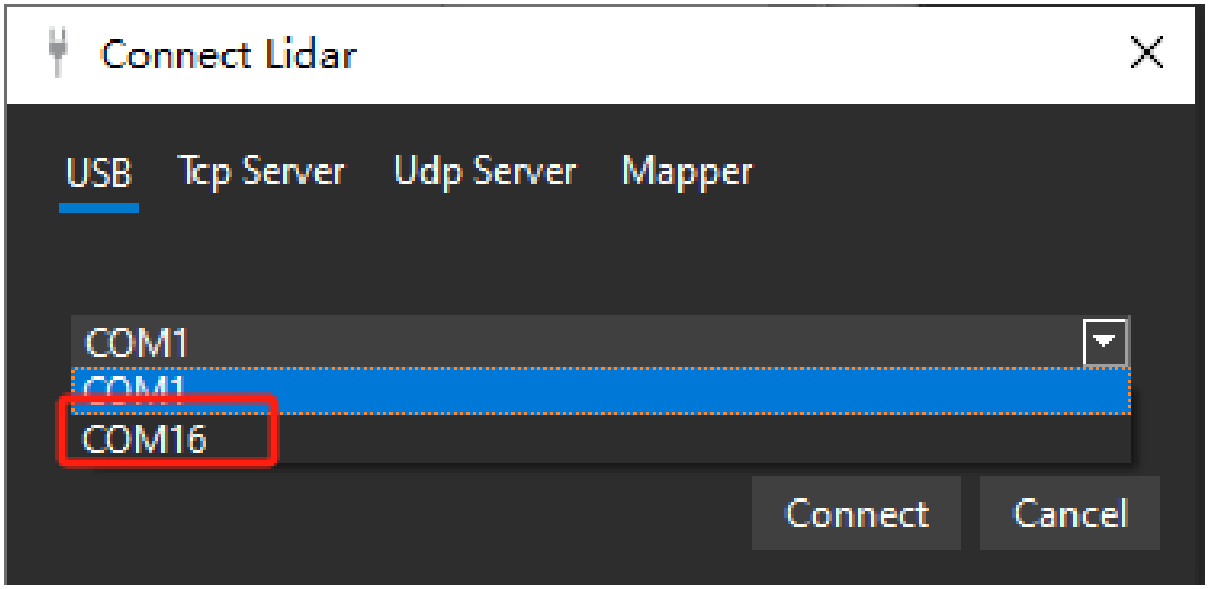
- After selecting, a pop-up window appears as shown below. Select the COM port identified earlier at the USB location, and after making the selection, click "Connect".



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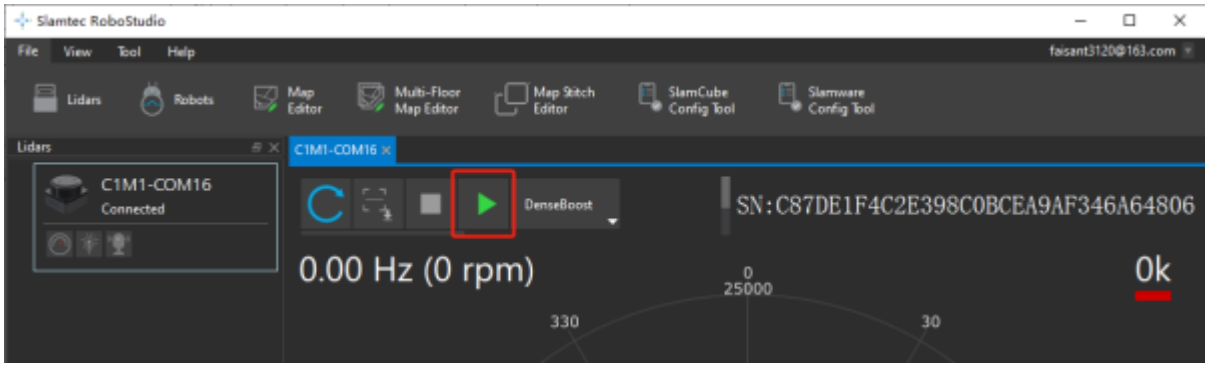
- After connecting, you will be able to see the radar in the window. Click on the triangle button in the top left corner, and the radar will begin scanning.



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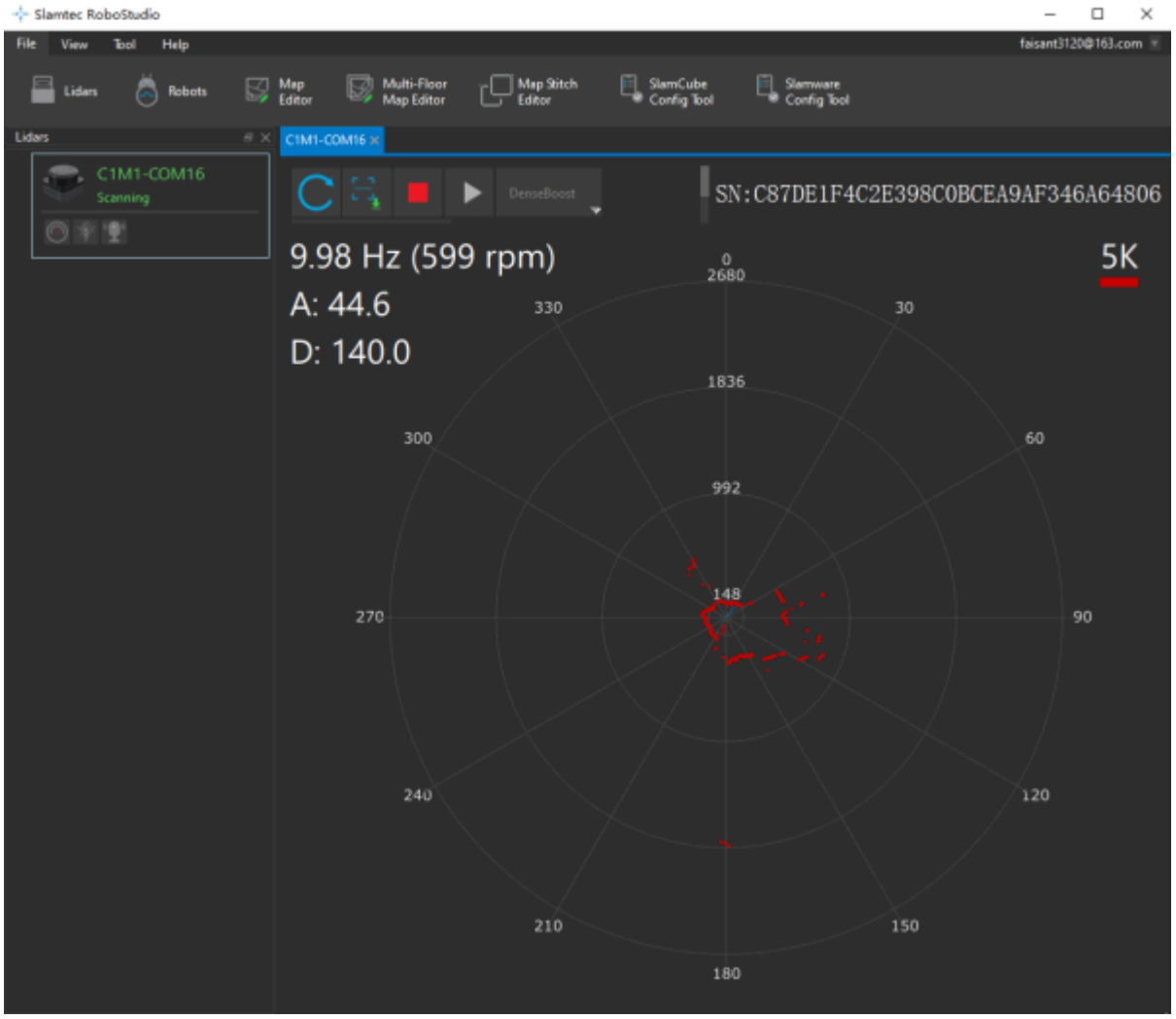
- Now, you can see the results of the radar scan on the screen.



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Resource

Document

- User Manual (https://files.waveshare.com/wiki/RPLIDAR-C1/SLAMTEC_rplidarkit_usermanual_C1_v1.0_en.pdf)
- Datasheet (https://files.waveshare.com/wiki/RPLIDAR-C1/SLAMTEC_rplidar_datasheet_C1_v1.0_en.pdf)

3D Model

- 3D Model (https://files.waveshare.com/wiki/RPLIDAR-C1/RPLIDAR_A1M8_(iges).zip)

Software

- SDK (https://files.waveshare.com/wiki/RPLIDAR-C1/Rplidar_sdk-master.zip)

- SDK address: https://github.com/Slamtec/rplidar_sdk (https://github.com/Slamtec/rplidar_sdk)

ROS Package

- ROS (https://files.waveshare.com/wiki/RPLIDAR-C1/Rplidar_ros-master.zip)
- ROS2 (https://files.waveshare.com/wiki/RPLIDAR-C1/Sllidar_ros2-main.zip)

Support

Technical Support

If you need technical support or have any feedback/review, please click the **Submit Now** button to submit a ticket, Our support team will check and reply to you within 1 to 2 working days. Please be patient as we make every effort to help you to resolve the issue.
Working Time: 9 AM - 6 PM GMT+8 (Monday to Friday)

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